Shear Amplitude Calculation in Abaqus Simulation

Parameter Definitions

- μ: Friction coefficient, unitless ($\mu = 0.70$)
- σ_n: Normal stress, in MPa ($\sigma_n = 10\,\text{MPa}$)
- Y: Yield strength of PMMA, in MPa ($Y = 3000\,\text{MPa}$)
- A: Contact area between spring and plate, $A = \texttt{SPR\_W} \times \texttt{SPR\_D}$, in mm^2
- h: Height of the spring (`SPR_H`), in mm
- D: Depth of contact interface (`DEPTH`), in mm
- L: Resistance area length, I use 220 mm

Step 1: Frictional Resistance Force

$$F_{\text{resist}} = 2\mu \cdot \sigma_n \cdot D \cdot L \tag{1}$$

Unit Check:

$$\begin{aligned}[\text{MPa}] \cdot [\text{mm}] \cdot [\text{mm}] &= \frac{\text{N}}{\text{mm}^2} \cdot \text{mm} \cdot \text{mm} \\ &= \text{N}\end{aligned}$$

So, F_{resist} has unit of N.

Step 2: Shear Amplitude

The shear amplitude is computed from:

$$\delta = \frac{F_{\text{resist}} \cdot h}{A \cdot Y} \tag{2}$$

Unit Check:

$$\begin{aligned}\frac{[\text{N}] \cdot [\text{mm}]}{[\text{mm}^2] \cdot [\text{MPa}]} &= \frac{\text{N} \cdot \text{mm}}{\text{mm}^2 \cdot \frac{\text{N}}{\text{mm}^2}} \\ &= \text{mm}\end{aligned}$$

Thus, δ (shear amplitude) has unit of mm, which is physically consistent as a displacement.

Alternative Expression (Code Form)

In Python-style code:

```
RESISTANCE = 2 * FRICTION_COEFFICIENT * DEPTH * 220 (mm) * NORMAL_STRESS
SHEAR_AMPLITUDE = RESISTANCE * SPR_H / ( SPR_W * SPR_D * Y_PMMA )
```

Or, more explicitly:

```
CONTACT_AREA = SPR_W * SPR_D
SPRING_STIFFNESS = Y_PMMA * CONTACT_AREA / SPR_H
SHEAR_AMPLITUDE = RESISTANCE / SPRING_STIFFNESS
```